

Customer Shopping Behavior Analysis

Uncovering insights from transactional data to guide strategic business decisions.





Project Overview

1

Data Source

3,900 purchases across product categories.

2

Key Goal

Uncover spending patterns, segments, preferences.

3

Outcome

Guide strategic business decisions.



Dataset Summary

- **Rows:** 3,900
- **Columns:** 18
- **Missing Data:** 37 values in Review Rating.

Key Features

- Customer demographics (Age, Gender, Location, Subscription Subscription)
- Purchase details (Item, Category, Amount, Season, Size, Color)
- Shopping behavior (Discount, Promo, Previous Purchases, Frequency, Review, Shipping)

Exploratory Data Analysis (Python)

01

Data Preparation

Loading, initial exploration, missing data handling (median imputation).

02

Column Standardization

Renamed to snake_case for readability.

03

Feature Engineering

Created age_group and purchase_frequency_days.

04

Data Consistency

Dropped redundant 'promo_code_used'.

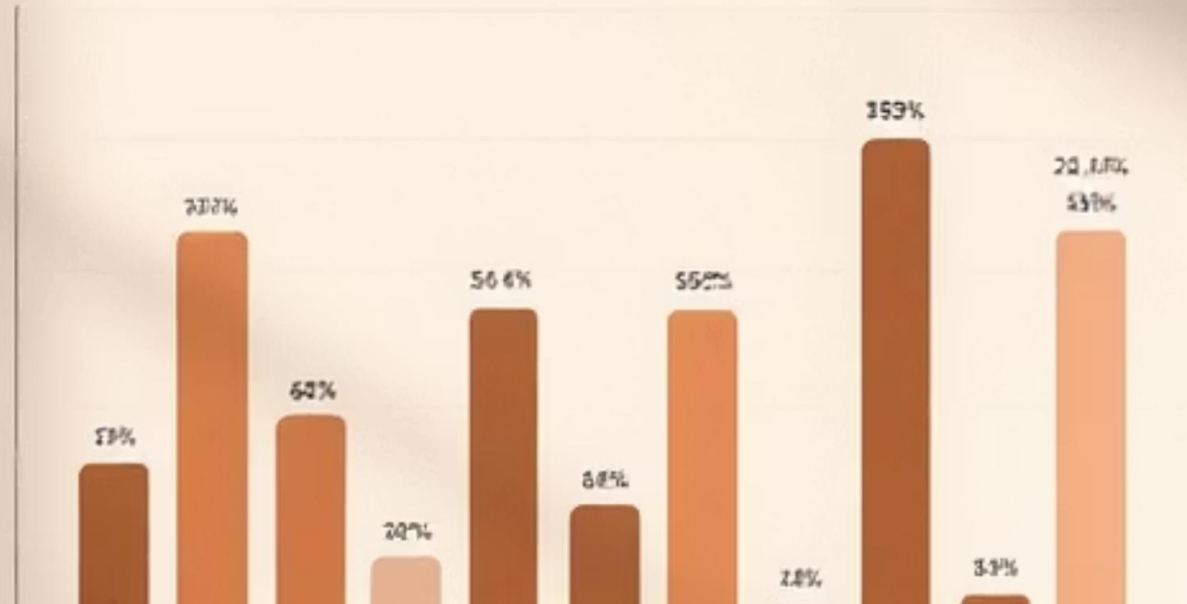
05

Database Integration

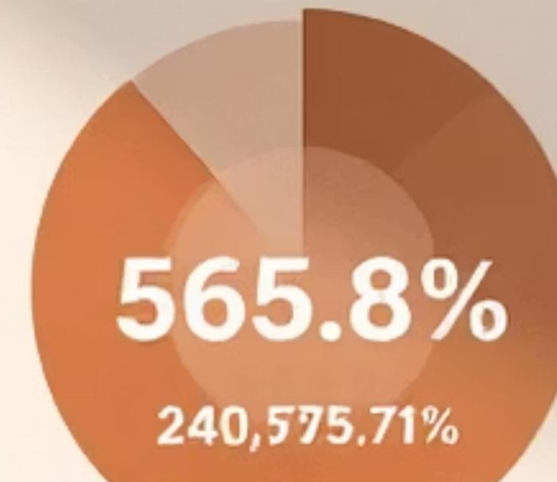
Loaded cleaned data to PostgreSQL for SQL analysis.

Revenue by Gender

Product Ratings



Product Ratings



Data Analysis Using SQL: Key Findings

Revenue by Gender

Male: \$157,890

Female: \$75,191

Top 5 Products by Rating

- Gloves (3.86)
- Sandals (3.84)
- Hat (3.81)
- Boots (3.81)
- T-shirt (3.78)



SQL Insights: Spending & Shipping

High-Spending Discount Users

Identified customers using discounts but spending above average.

- Customer 2: \$64
- Customer 3: \$73
- Customer 4: \$90

Shipping Type Comparison

Express: Avg. \$60.48

Standard: Avg. \$58.46



SQL Insights: Subscriptions & Discounts

Subscribers vs. Non-Subscribers

Subscribers: Avg. \$59.49, Total \$62,645

Non-Subscribers: Avg. \$59.87, Total \$170,436

Discount-Dependent Products

- Hat (50.00%)
- Sneakers (49.66%)
- Coat (49.07%)



SQL Insights: Segmentation & Age

Customer Segmentation

- **Loyal:** 3116 customers
- **Returning:** 701 customers
- **New:** 83 customers

Revenue by Age Group

- **Young Adult:** \$62,143
- **Middle Age:** \$59,197
- **Adult:** \$55,978
- **Senior:** \$55,763

Customer Behavior Dashboard

subscription Status

No Yes

Gender

Fem... Male

Category

Accessories
Clothing
Footwear
Outerwear

Shipping Type

☐ 2-Day Shipping
☐ Express
☐ Free Shipping
☐ Next Day Air
☐ Standard
☐ Store Pickup

3.9K

number of customers

\$59.76

Average purchase amount

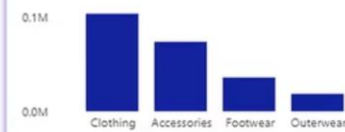
3.75

average review rating

% of Customers By Subscription Status



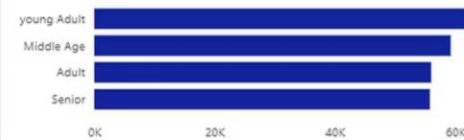
Revenue By Category



Sales By Category



Revenue By Age Group



Sales by Age Group



Dashboard Analysis (Power BI)

Visualizing key metrics: customer count, average purchase, purchase, review rating, and revenue by category/age.

Business Recommendations



Boost Subscriptions

Promote exclusive benefits.
benefits.



Customer Loyalty

Reward repeat buyers.



Review Discount Policy

Balance sales with margin.
margin.



Product Positioning Positioning

Highlight top-rated items.
items.



Targeted Marketing

Focus on high-revenue groups.